

Mohamed Suhaib A

+91 6369818684 mohamedsuhaib167@gmail.com LinkedIn GitHub

SUMMARY

Electronics and Communication Engineering student with hands-on experience in Embedded Systems, IoT, AI, and Machine Learning through internships and academic research projects. Proficient in C, Python, ARM programming, and hardware-software integration, with a strong foundation in real-time system development and problem-solving. Passionate about building innovative engineering solutions and eager to contribute to embedded, electronics, and software roles.

EXPERIENCE

Embedded Systems Intern

May 2025 – Jun 2025

Emertxe Information Technologies

Remote

- Completed industry-oriented training focused on embedded systems development and microcontroller-based applications.
- Developed embedded C programs involving GPIO, timers, interrupts, and peripheral interfacing concepts while following embedded software development workflows.
- Gained practical experience in firmware development, debugging, and hardware-software integration for embedded systems.

Industrial Intern

Jun 2024

ACE Electrotrend Panels Pvt. Ltd.

Hosur, Tamil Nadu

- Undertook industrial training focused on electrical panel manufacturing and automation systems.
- Studied electrical schematics, panel wiring layouts, assembly processes, and industrial control system workflows.
- Developed an understanding of industrial electrical systems, automation practices, and engineering documentation standards.

TECHNICAL SKILLS

Programming: C, Python

Embedded Systems: ESP32, Arduino, GPIO, Timers, Interrupts

Communication Protocols: UART, SPI, I2C

Computer Vision: OpenCV, YOLOv8

VLSI Tools: Cadence Virtuoso

Tools: Git, GitHub, Linux, VS Code

PROJECTS

AI-Powered Threat Detection and Surveillance System | *Python, OpenCV, YOLOv8*

- Developed a computer vision-based surveillance system for real-time object monitoring and threat detection.
- Implemented object detection and tracking using YOLOv8 and OpenCV and integrated video processing modules into a unified platform.
- Demonstrated intelligent surveillance capabilities through automated event detection and monitoring.

AI-Powered Traffic Management System | *Python, OpenCV, YOLOv8*

- Developed a smart traffic monitoring system for vehicle density analysis and adaptive signal control.
- Implemented vehicle detection and counting using YOLOv8 and OpenCV for traffic flow monitoring.
- Demonstrated adaptive traffic management concepts based on real-time traffic conditions.

EEG-Based Eye Blink Detection System | *Python*

- Developed an EEG-based eye blink detection system for Brain-Computer Interface (BCI) applications.
- Performed signal preprocessing, feature extraction, and classification of EEG signals using Python.
- Successfully identified eye blink patterns from EEG data and gained exposure to biomedical signal analysis.

CMOS LC Voltage Controlled Oscillator | *Cadence Virtuoso*

- Designed and simulated a CMOS LC Voltage Controlled Oscillator (LC-VCO) for analog/RF applications.
- Performed transient and frequency-domain simulations to analyze tuning range and phase-noise characteristics.
- Gained hands-on experience in CMOS analog circuit design, RF building blocks, and circuit-level simulation techniques.

EDUCATION

SRM University AP

2023 – 2027

B.Tech in Electronics and Communication Engineering

CGPA: 8.38/10

CERTIFICATIONS

Google Cybersecurity Professional Certificate

Google / Coursera

Dec 2024

TCS iON Career Edge – Young Professional

TCS iON

Jul 2024